

When Affective Cues Disagree: Counterfactual Twins for Auditing Visual Emotion Models

Motivation. Visual emotion models are usually judged by whether their predictions match image-level annotations. This is insufficient: a model can return the expected emotion because it follows colour, text, or background correlations, while its explanation names a face. Recent VLM benchmarks reveal errors in affective reasoning and intensity estimation [1], but output quality alone cannot establish which visual evidence determined a judgement. We ask a stricter question: *if one affective cue changes while the visible event remains otherwise stable, does the model revise its prediction, explanation, and uncertainty for the right reason?*

Counterfactual Affective Twins. We propose a diagnostic benchmark of controlled original-counterfactual pairs. Each “twin” changes one inspectable cue family—an AU-related facial region, colour and lighting, scene context, or embedded text—and records the edit mask and preserved content. A model passes only when four behaviours agree: it responds to the changed evidence, remains invariant to what was preserved, grounds its explanation in the manipulated region, and reduces confidence when cues genuinely conflict. These tests separate *cue responsiveness*, *selective invariance*, *evidence grounding*, and *conflict calibration*, which ordinary accuracy conflates. The model jointly reports categorical emotion, valence-arousal, visible evidence, a literal caption, and uncertainty. Viewer-response distributions are retained rather than treating one label as an objective emotional truth.

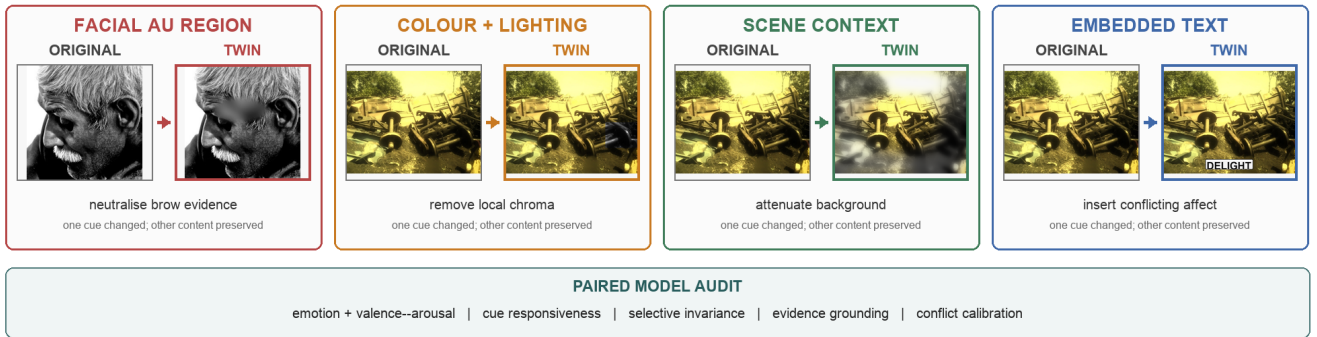


Figure 1: Representative outputs from the implemented intervention generator. Quantitative evidence is reported for colour/lighting; AU-related face, context, and text pairs illustrate the full audit design pending human validation.

Feasibility study. We first test a necessary safeguard: does an affect-sensitive intervention selected by one model transfer to an independently fitted evaluator, or merely exploit the locator? Emotion6 contains 1,980 images across six categories and retains viewer-response distributions [2]. Using frozen ImageNet-pretrained ResNet-18 features [3], we train two emotion heads with independent initialisation and bootstrap samples; the evaluator obtains 51.7% accuracy and 0.512 macro-F1 on the 402-image test split. For 60 balanced held-out images, the locator ranks 16 equal-area regions by the reduction in annotated-class probability after luminance-preserving chroma removal. The evaluator then assesses the selected region against all 15 area-matched controls. Selected interventions reduce annotated-class probability by 0.070, compared with -0.001 for controls. The paired advantage is 0.071 (95% bootstrap CI [0.050, 0.096]) and is positive for 58 of 60 images. Thus, the detected colour reliance transfers across independently fitted heads rather than remaining a locator-specific saliency artefact.

From feasibility to benchmark. The executable prototype now produces twins, masks, and intervention records for all four cue families. Human raters will next verify edit naturalness, target-cue change, and non-target preservation before the pairs are treated as perceptual counterfactuals. We will then compare emotion classifiers and VLMs through a cue-conflict matrix designed to expose failures hidden by aggregate accuracy: which evidence dominates, whether explanations follow the edited region, and whether uncertainty rises under genuine ambiguity. The intended contribution is not another emotion leaderboard, but an auditable account of what visual emotion models use, ignore, and merely mention. This provides a common diagnostic layer across facial behaviour, dimensional affect, scene emotion, and emotion-aware captioning, asking whether affective evidence is structured, transferable, and inspectable.

References. [1] D. She et al. “AICA-Bench: Holistically Examining the Capabilities of VLMs in Affective Image Content Analysis.” *Findings of ACL*, 2026. [2] K.-C. Peng et al. “A Mixed Bag of Emotions: Model, Predict, and Transfer Emotion Distributions.” *CVPR*, 2015. [3] K. He et al. “Deep Residual Learning for Image Recognition.” *CVPR*, 2016.